Publishability Assessment: MPM Extension of A-Shaped Mat Foundation Study

Critical opportunity identified with specific requirements for success

Bottom line: Extending Liu et al.'s (2022) A-shaped mat foundation study using MPM represents a **high-impact publication opportunity** with multiple novel contributions, but success requires going beyond simple FEM validation. The research fills a critical gap—**zero MPM applications exist for mat foundations of any geometry**—and Liu et al.'s work has major limitations (static loading only, small-strain FEM, no installation effects). However, **purely replicating FEM results is insufficient for top journals**; you must demonstrate unique MPM capabilities like installation effects, large deformation mechanisms, or post-peak behavior.

This assessment is based on analysis of 100+ recent papers (2020-2025) from target journals, revealing that successful FEM-MPM comparison papers combine methodological innovation with case history validation and practical insights. The A-shaped mat foundation domain has substantial unexplored territory, and MPM's proven advantages in offshore geotechnics (spudcans, pipelines) position it well for this application—if approached strategically.

The critical research gap: Mat foundations completely absent from MPM literature

MPM applications in offshore geotechnics have exploded from 2020-2025, but with a glaring omission. (ScienceDirect +10) **Spudcan penetration** dominates the field with commercial applications emerging (Beuckelaers 2024 developed MPM-based apps now progressing toward industry standard). **Pipeline-soil interaction** has 6+ major papers showing MPM capturing lateral movement, uplift, and penetration mechanisms. **Suction caissons** have limited but growing attention with 2-3 studies demonstrating promise for installation analysis.

Yet mat foundations—despite experiencing identical large deformation physics during installation—have **zero MPM applications**. Not for conventional rectangular mats. Not for A-shaped mudmats common in jacket installations. Not for perforated or multi-compartment designs. The entire foundation class remains unexplored despite MPM's demonstrated success with analogous penetration problems like CPT simulation, where Martinelli & Galavi (2021) and Bisht & Salgado (2024) showed MPM handles "very large displacement levels" where FEM fails due to mesh distortion.

This gap exists for A-shaped mats specifically. Industry sources confirm these foundations are "typically A-shaped" for offshore jacket temporary support, designed with "non-conventional geometry and internal skirts" requiring "3D FEA analysis as valuable addition to conventional design methods"—but **exclusively using FEM, never MPM**. (ResearchGate) (ResearchGate) The complex geometry with perforations, varying skirt depths, and internal stiffeners creates precisely the stress concentrations, soil flow patterns, and large deformations where MPM excels over FEM. (Number Analytics)

Why this matters for publishability: First-time application of an established method to an important problem class can be highly publishable if executed well. The mat foundation gap is not due to unsuitability—the physics match MPM's strengths perfectly—but rather community separation between MPM developers and foundation designers.

Liu et al.'s limitations create multiple extension opportunities

Liu et al.'s (2022) study in Ships and Offshore Structures established baseline bearing capacity for A-shaped mats using FEM plus centrifuge validation, examining vertical, horizontal, and moment capacity with V-H-M failure envelopes for two interface conditions. **But the limitations are extensive:**

Loading conditions were purely static and monotonic. No cyclic wave loading despite continuous ocean action on offshore platforms. No dynamic earthquake effects. No rate-dependent analysis even though offshore foundation literature extensively documents strain rate affecting undrained strength by 10-25% per log cycle—critical for differentiating fast installation from slow operational loading. Studies by Zhu et al. (2022-2023) on conventional mats show cyclic loading causes 20-40% capacity reduction, but A-shaped geometry effects remain unknown.

Soil complexity was minimal: homogeneous cohesive soil only. Reality presents stratified clay layers, sand-over-clay sequences, strength gradients with depth. Consolidation effects completely ignored despite Zhang et al. (2022) and Long et al. (2022-2023) showing consolidation increases capacity 30-100% for general mudmats. Strain softening and soil degradation—critical for A-shaped stress concentrations—not captured in small-strain FEM. No consideration of natural clay structure or over-consolidation ratio effects. ScienceDirect

Installation and operational effects entirely absent. No soil disturbance modeling during placement. No heave mound formation. No skirt penetration resistance. No remolding or excess pore pressure generation that changes subsequent bearing capacity. ScienceDirect Scour effects around mat edges completely unexplored despite being capacity-critical for offshore foundations. ScienceDirect

Numerical methodology used small-strain FEM incapable of modeling large deformations, progressive failure, or post-peak behavior. These limitations align precisely with MPM's documented advantages, MDPI creating natural extension opportunities.

What FEM-MPM comparison papers must deliver for publication

Analysis of recent comparison studies reveals **purely numerical comparisons are insufficient** for top journals. The pattern across successful papers from 2020-2025 is consistent and demanding.

Top-tier journals (JGGE, Géotechnique, Acta Geotechnica) require case history or extensive experimental validation—not just FEM-MPM agreement. ASCE Library Sordo et al.'s (2024) JGGE paper on tailings dam failures succeeded by simulating the 1978 Mochikoshi dam with real earthquake data, demonstrating strain-softening beyond liquefaction is required for accurate runout. Lu et al.'s (2023) Acta Geotechnica paper integrated FEM-MPM with machine learning for probabilistic landslide risk, providing practical tools for exceedance probability quantification. (Springer) Both went far beyond comparison to deliver novel frameworks with physical insights.

The critical finding: **validation against FEM alone is not publishable** in top venues. Yost et al.'s (2022) JGGE paper on MPM for CPT succeeded through extensive calibration chamber validation in layered soils, quantifying thin-layer effects with errors under 5-10%. (ASCE Library) Physical experiments or documented case histories provide the credibility pure numerical comparison lacks.

Computers and Geotechnics accepts comparison papers but with strict requirements for algorithmic novelty. (ScienceDirect) The journal explicitly discourages papers that "predominantly report results from proprietary codes" or "describe computer modelling of laboratory tests" unless demonstrating "novel user-implemented computational methods." (ScienceDirect +2) Recent accepted papers show the bar: Zhang et al. (2022) developed a hybrid contact method combining point-point and point-segment advantages with 3D implementation and multiple benchmark validations (errors under 6%). Huo et al.'s (2025) GPU-accelerated MPM achieved 21.3× speedup over multi-threaded CPU with 67% memory bandwidth utilization, handling 190M particles. (ScienceDirect)

Pattern across all successful papers: Novel methodology + rigorous validation + practical insights. Fernández et al. (2024) compared FEM, Numerical Limit Analysis, and MPM for shaft excavation, introducing 3D strength reduction factor methodology within MPM and demonstrating strategic method combinations. (Springer) Ceccato et al. (2023) validated sequential FEM-to-MPM mapping against laboratory experiments using digital image correlation, addressing computational efficiency for rainfall-induced landslides. (Springer)

What does NOT make it publishable: Simple demonstrations that "MPM handles large deformation better than FEM" (already well-known). Comparisons on hypothetical problems without validation. Agreement between methods without experimental data. Application of existing codes without methodological innovation. Single benchmark problems without practical implications.

MPM's documented advantages perfectly match mat foundation failure mechanisms

The literature establishes where MPM excels—and these align precisely with A-shaped mat foundation physics. (ScienceDirect +3) **Progressive failure and strain localization** are captured naturally in MPM through correctly oriented shear bands without mesh dependency. (Number Analytics +3) Martinelli & Remmerswaal (2023) showed MPM handles post-peak strain softening with progressive strength degradation. (ISSMGE) Strip footing studies demonstrate MPM captures non-crystallographic shear bands in dense sand while FEM requires artificial remeshing. (Sage Journals) (Taylor & Francis)

Penetration problems—the closest analogue to foundation bearing capacity—provide the strongest evidence. CPT literature documents FEM failures: Susila & Hryciw (2003) required "auto-adaptive remeshing for very large distortion" with accuracy losses. Fan et al. (2018) found "several issues related to accuracy and stability" using ALE in Abaqus. (ScienceDirect) Multiple studies note FEM's "numerically tiresome remeshing and mapping of state variables." (University of Dundee) (ScienceDirect) In contrast, Bisht & Salgado (2024) state "It is now possible to realistically simulate cone penetration in either sand or clay using MPM" with excellent agreement against DIC analysis, concluding: "Because the grid can always be reset between computation time increments, mesh distortion does not impact calculations, and MPM simulations can progress to very large displacement levels."

Post-peak and post-failure behavior differentiates the methods fundamentally. Direct comparisons show both methods consistent in small strain regimes, but "in large strain deformation, MPM is still able to get convergent result while FEM is not" with MPM animating post-failure behavior, calculating stress distributions, and presenting final geometry (Journal of Civil Engineering Forum, 2023). Standard Lagrangian FEM suffers "mesh entanglement making post-failure analysis considerably challenging if not impossible" (MDPI) (Al-Kafaji 2013). (ScienceDirect)

Contact and interface modeling comes at "no extra cost with standard MPM formulation" for no-slip, no-penetration contact. (ScienceDirect) (ScienceDirect) Frictional contact implements easily through background grid without special elements. (Albandevaucorbeil) Multiple contact bodies handle naturally through multi-velocity field techniques. (ResearchGate) (ScienceDirect) Ma et al. developed geo-contact algorithms specifically for geotechnical applications, successfully applied to foundation penetration and soil-structure interaction without artificial penetration. (ScienceDirect)

Soil flow and material movement—critical for A-shaped mats with perforations—occur naturally as material points carry history-dependent information through fixed Eulerian grids. Soil flows freely without mesh constraints, naturally handling topology changes like flow around foundations or through perforations. (Wikipedia) (Albandevaucorbeil) This capability directly addresses A-shaped mat design questions: How do perforations affect soil plug formation? What are installation effects for complex geometry? How does soil flow around internal configurations?

The critical deformation threshold where MPM advantages become essential: literature suggests displacements exceeding 10-20% of foundation width, (ScienceDirect) post-peak behavior analysis, strain localization, progressive failure, or complex contact conditions. **A-shaped mats with perforations likely experience all these conditions during installation and failure.**

V-H-M failure envelope modeling: Unexplored territory for MPM

This represents perhaps the most significant gap. While V-H-M (Vertical-Horizontal-Moment) failure envelopes are well-established for offshore foundations using FEM/FELA methods— (Wiley Online Library)with seminal works by Gourvenec (2007), Bransby & Randolph (1998), and comprehensive methodology comparisons by Suryasentana et al. (2020)—** (strath)extensive searches found zero evidence of MPM being used to generate V-H-M failure envelopes.**

The methodologies exist: displacement probe tests applying prescribed ratios in different directions, load probe tests using FELA, swipe tests holding one load constant while increasing another, and sequential swipe methods with incremental loading. (strath) These approaches map full envelopes typically requiring 10-100 simulations depending on resolution. (ScienceDirect) (Strathprints) The standard numerical approach is FEM-dominated (~95%

of published work) with small deformation assumptions and wished-in-place analysis ignoring installation effects. Wiley Online Library

Why MPM hasn't been applied likely involves computational cost (MPM 5-10× slower requiring weeks to months for full 3D envelopes), methodology transfer challenges (probe/swipe methods designed for FEM convergence behavior), community separation (MPM researchers focus on extreme events while foundation engineers use established tools), and validation requirements against extensive FEM/centrifuge databases.

But the opportunities are substantial. MPM could deliver **post-peak and residual capacity envelopes** where FEM shows peak capacity only, enabling understanding of progressive failure under storm loading with potential 30-50% residual capacity. **Installation-affected envelopes** could capture soil remoulding and strength degradation—current practice uses wished-in-place analysis, but CEL studies for uniaxial loads show post-installation capacity 20-40% lower. ScienceDirect **Large-displacement load path effects** could reveal envelope evolution as foundations move and rotate, producing "dynamic failure envelopes" showing capacity reduction with accumulated displacement.

Additional novel contributions include scour-affected capacity with time-varying envelopes for life assessment, multi-event loading sequences showing evolved envelopes after N cycles, and failure mechanism visualization tracking soil flow and failure surface evolution through large displacement revealing transitions invisible to FEM.

Critical challenges include load path dependency at large deformation where envelope shape may depend on deformation magnitude and different paths produce different capacities. Defining "failure" in MPM requires criteria like displacement thresholds or load plateaus rather than FEM's clear convergence failure. Computational cost for full envelope mapping with 50-100+ simulations could require weeks to months. Implementing probe/swipe methods in MPM needs quasi-static formulations to prevent load drift during "holding" phases. ScienceDirect

Making this publishable requires going beyond FEM replication. Demonstrating unique capabilities—installation-affected envelopes with highest practical impact, post-peak envelopes for safety-critical applications, large displacement evolution, or failure mechanism transitions. Rigorous validation against both FEM benchmarks (like Gourvenec 2007 circular footing data) and centrifuge tests. Practical design outcomes with correction factors, design charts, or quantified capacity reduction versus displacement. This represents 18-30 month effort but with **Géotechnique or JGGE publication potential** given the methodological advance plus practical implications.

Computational efficiency claims require rigorous benchmarking

Recent MPM papers in Computers and Geotechnics set high standards for efficiency claims. Huo et al. (2025) achieved **21.3× GPU speedup** over multi-threaded CPU with detailed memory bandwidth analysis showing 67% average utilization (78% peak). The paper provided complete hardware specifications (GH200 platform, 20-core CPU comparison), timing breakdowns by function, and demonstrated 190M particle capacity. Wyser et al. (2020) reported **28-50× speedup** through vectorization with sustained 600-900 Mflops performance, identifying L2 cache thresholds at ~1000 material points for peak performance.

Minimum publishable thresholds vary by improvement type: algorithmic improvements need 2-5× speedup, implementation optimization 10-20×, hardware acceleration 20-100×, combined multi-GPU approaches 100-1000×. But **quality matters more than magnitude**—reproducibility with clear specifications, fair comparison at same accuracy levels, scaling analysis versus problem size, breakdown analysis of time spent in each function, and practical validation on realistic problems.

For FEM-MPM comparisons specifically, the pattern shows MPM generally more expensive due to particle-grid mapping overhead but maintains convergence where FEM fails. Small deformation favors FEM for efficiency and accuracy. Large deformation makes MPM necessary despite higher cost—**the practical threshold occurs when strain exceeds 20-30%** where mesh distortion prevents FEM convergence. Tudelft

What this means for your study: Don't claim MPM is "more efficient" than FEM for bearing capacity generally—it's typically slower for small deformations. Instead, frame efficiency correctly: (1) MPM enables analyses FEM cannot complete due to convergence failure, (2) For large deformation problems, MPM avoids expensive remeshing cycles, (3) Installation-to-operation sequences require only one MPM model versus separate FEM analyses, (4) Computational cost justified by unique insights unavailable from FEM.

Report metrics rigorously: wall-clock time with hardware specs (CPU model, cores, frequency, RAM), time per analysis phase (mapping, solving, updating), convergence behavior compared to FEM, memory usage, and number of increments required. Compare total computational cost for equivalent accuracy rather than per-iteration speed.

Publication strategy: Target journals and required scope

Tier 1 targets require major novel contributions. **Computers and Geotechnics** (IF: 5.3-6.7) fits best with emphasis on numerical algorithms and accepting comparison papers if showing methodological innovation. Resurchify +4 Required elements: Novel MPM implementation for mat foundations (first application), 3-5 benchmark validations showing accuracy, computational efficiency analysis, at least one practical A-shaped mat application, parametric study demonstrating method capabilities. Estimated scope: 12-15 pages, 12-15 figures, 4-6 month effort post-FEM validation.

Ships and Offshore Structures published Liu et al.'s original work and explicitly covers "offshore platforms, jack-ups, and FPSOs." An MPM extension would fit the journal's scope perfectly, especially if emphasizing practical implications for jack-up mudmat design. Slightly lower bar than C&G for algorithmic novelty but requires offshore engineering relevance. Could frame as "large deformation installation effects on A-shaped mat capacity—implications for jack-up platform safety."

Ocean Engineering or **Applied Ocean Research** accept offshore foundation papers with emphasis on practical applications. These venues would value installation effects, cyclic loading analysis, or scour effects even with less algorithmic novelty than C&G demands.

Stretch target: Géotechnique (IF: ~4.5) if pursuing V-H-M envelope generation with MPM. Requires rigorous experimental validation, comparison with established Gourvenec-type benchmarks, physical insights into failure mechanisms, and demonstration of when/why MPM-predicted envelopes differ from conventional FEM approaches. This represents 18-30 month effort but offers highest impact if successful.

Publication pathway recommendation: Start with Computers and Geotechnics targeting 3-4 specific novel contributions from the prioritized list below. After establishing MPM credibility for mat foundations, pursue Géotechnique with V-H-M envelope methodology paper. This two-paper strategy maximizes impact while managing risk.

Recommended novel contributions ranked by publishability

Tier 1 (Essential—include at least 2 for top journal publication):

Installation effects on bearing capacity stands as highest impact. Model spudcan/mat penetration using MPM capturing soil remoulding, heave mound formation, and changed in-situ conditions, then analyze post-installation bearing capacity. ResearchGate +2 Compare with Liu et al.'s wished-in-place FEM results. Expected finding: 20-40% capacity reduction (based on CEL studies for other foundation types). ScienceDirect Unique to MPM—FEM cannot complete full installation-to-bearing sequence without remeshing errors. Wikipedia Practical implications: safety factors in current design may be inadequate. This contribution alone could carry a paper in Ocean Engineering or Ships and Offshore Structures.

Rate-dependent behavior during installation versus operation addresses documented but unexplored strain rate effects. Implement viscoplastic soil model capturing 10-25% strength change per log cycle of strain rate. Simulate fast installation (minutes to hours) versus slow operational loading (days to years). Show capacity differences between rate-dependent and conventional analyses. First study linking installation rate to A-shaped mat capacity. Practical value: optimal installation speed, time-dependent capacity evolution. Strong contribution for Computers and Geotechnics with proper constitutive model implementation.

Large deformation failure mechanisms with complex geometry leverages MPM's visualization capabilities. Track soil flow around A-shaped perforations, identify stress concentrations at geometry discontinuities, map progressive failure initiation zones, demonstrate failure mechanism transitions invisible to small-strain FEM. Number Analytics Cfms-sols High-quality figures showing particle trajectories, evolving failure surfaces, and strain localization patterns. Physical insights justifying computational investment. Essential for demonstrating why MPM was necessary—cannot just show it works, must show what FEM misses.

Tier 2 (Strong additions—include 1-2 for comprehensive paper):

Cyclic loading with progressive settlement models wave loading cycles with accumulated plastic deformation, progressive foundation settlement, and capacity degradation with load cycles. Compare with Zhu et al.'s work on conventional mats showing 20-40% reduction, determine A-shaped geometry effects. Addresses real operational condition ignored by Liu et al. Two-phase MPM formulation may be needed for pore pressure effects.

Consolidation effects on long-term capacity uses two-phase MPM for pore pressure dissipation post-installation, tracks partially drained behavior, quantifies capacity gain with time (potentially 30-100% based on literature). ScienceDirect Demonstrates full installation-consolidation-operation sequence in unified MPM framework. Strong practical contribution for operational capacity assessment.

Strain softening in irregular stress distributions specific to A-shaped geometry where stress concentrations at corners and perforations cause localized softening, progressive failure propagation, and sensitivity loss. Cfms-sols Hindawi Shows MPM captures effects impossible in elastic-perfectly plastic FEM. Good for physical insight and mechanism understanding.

Tier 3 (Valuable but not essential for first paper):

Heterogeneous/layered soil effects, scour pit modeling, parametric design charts varying aspect ratio and soil properties, comparison with centrifuge data (if available), dynamic loading for earthquake analysis. These could form follow-up papers or extended versions.

Critical success factors and potential pitfalls

Success requires strategic execution beyond technical competence. The research gap is real and significant, but approaches matter enormously for publishability.

Essential requirements: Validate against Liu et al.'s FEM results first—reproduce their baseline cases showing MPM accuracy in small deformation regime before claiming advantages. Choose 2-3 Tier 1 novel contributions and execute them thoroughly rather than superficial treatment of many aspects. Provide rigorous validation with quantitative error metrics (target under 10% for benchmark cases). Focus on physical insights and practical implications in every result section—explain why findings matter for design, safety, or understanding.

Common pitfalls to avoid: Claiming MPM is "better" than FEM generally—it's not, it's different with specific advantages. Insufficient validation—comparison with only Liu et al. without additional benchmarks. Purely demonstrative work without novel insights—showing MPM works is insufficient. Computational cost without justification—must demonstrate insights justify expense. Overpromising on efficiency—MPM typically slower per analysis than FEM for problems within FEM's convergence range.

Validation strategy recommendation: Reproduce 2-3 of Liu et al.'s load cases showing MPM accuracy within 10%. Validate against at least one additional independent source (centrifuge data if available, or classical bearing capacity problems with known solutions). Perform convergence studies demonstrating grid independence. Compare failure mechanisms with established patterns from bearing capacity theory.

Realistic timeline for high-quality publication: FEM validation and benchmarking 3-4 months. Implementation of novel contribution 1 (installation effects) 4-6 months. Implementation of novel contribution 2 (rate effects or large deformation mechanisms) 3-4 months. Paper writing, figure generation, revision 2-3 months. Total: 12-17 months from starting MPM development to submission. Factor additional 3-6 months for review and revision. Papers attempting too much too quickly often fail—depth beats breadth for top journals.

Conclusion: High-impact opportunity with clear execution path

This research extension is **highly publishable if executed strategically**. The convergence of factors favors success: complete absence of MPM for mat foundations despite proven success in analogous offshore applications, Liu et al.'s significant limitations creating multiple extension opportunities, well-documented MPM advantages for large deformation precisely matching mat foundation failure physics, MDPI and V-H-M envelope generation representing completely unexplored territory. Sage Journals +3

The opportunity is not merely incremental—it's transformative for this foundation class. First MPM application to any mat foundation geometry. First large deformation installation analysis of A-shaped mats. First rate-effects study for this geometry. First cyclic loading analysis capturing accumulation effects. These "firsts" carry weight, but **only if accompanied by physical insights, practical implications, and rigorous validation**.

Your strategic advantage comes from Liu et al.'s centrifuge data providing validation targets and FEM results showing what conventional analysis predicts. ScienceDirect You can systematically demonstrate where conventional approaches remain valid (small deformations, wished-in-place) and where MPM reveals critical effects (installation, large displacement, progressive failure). This positions MPM as complementary advancement rather than replacement—increasing acceptance likelihood.

Recommend proceeding with focus on 2-3 Tier 1 contributions (installation effects essential, plus rate effects or large deformation mechanisms). Target Computers and Geotechnics initially with 12-15 month timeline. If V-H-M envelope generation interests you, pursue as second paper targeting Géotechnique with 18-24 month effort. The research fills genuine needs in offshore foundation design where jack-up platforms, subsea structures, and temporary supports all rely on mudmat foundations experiencing large deformations that current small-strain methods cannot properly characterize. ResearchGate +7

Success probability is high with proper execution. The technical barriers are surmountable—MPM software exists (Anura3D, CB-Geo MPM), constitutive models are available, and computational resources are accessible. ScienceDirect ScienceDirect The scientific barriers favor you—genuine gap, documented need, proven method advantages. The publication barriers are clear—follow patterns from successful comparison papers emphasizing novelty, validation, and insight rather than pure demonstration. **Proceed with confidence but strategic focus.**